

Question: Determine the most suitable bearing state type from...

Ground Truth: Undesired Input



...

Token Energy-Based Uncertainty Quantification

... description \” Moder ate Inner ...

$$(\cdots \text{NLL}_{\delta-2} + \text{NLL}_{\delta-1} + \text{NLL}_{\delta} + \text{NLL}_{\delta+1} + \text{NLL}_{\delta+2} \cdots) / N_t$$

Uncertainty score falls within the predefined margin



Acceptable uncertainty, return generation



Token Energy-Based Uncertainty Quantification

... description \” Moder ate Inner ...

$$\left(\begin{array}{cccccc} \cdots \text{NLL}_{\delta-2} & \text{NLL}_{\delta-1} & \text{NLL}_{\delta} & \text{NLL}_{\delta+1} & \text{NLL}_{\delta+2} & \cdots \\ \times & + & \times & + & \times & + & \times & + & \times \end{array} \right) / N_t$$
$$\left[1 + (\omega r_{\text{pb}} - 1) \mathbf{1}_{\{k=\delta-1\}} \right]$$

Uncertainty score is beyond the predefined margin



Unacceptable uncertainty, refuse to answer

